

Chapter 4

Elasticity, Revenue, and Consumer Surplus

“The rate at which one thing can be substituted for another is the most fundamental concept in economics.”

— Paul Samuelson, *Economics*, 1948

Before You Begin: Quick Self-Assessment

Test yourself on these prerequisites before reading:

1. **Intuitively, what does “elasticity” measure?** *How responsive quantity demanded is to a change in price (or income, or another good’s price).*
2. **If demand is elastic and a firm raises its price, what happens to total revenue?** *Total revenue falls — the percentage drop in quantity exceeds the percentage rise in price.*
3. **Draw a consumer surplus triangle on a linear demand curve.** *CS = area between the demand curve and the price line, from $Q = 0$ to Q^* . For linear demand: $CS = \frac{1}{2}(P_{\max} - P^*) \cdot Q^*$.*
4. **What is the cross-price elasticity of two goods that are complements?** *Negative — when the price of one rises, demand for the complement falls.*
5. **Name two factors that make demand more elastic.** *More available substitutes; good is a luxury rather than a necessity.*

Connecting Week 3 to Week 4

In Week 3 you derived the individual demand curve by tracing how the optimal bundle changes as P_x varies. You saw that demand slopes downward for normal goods, can shift with income, and decomposes into substitution and income effects. **This week we ask: by how much does quantity demanded respond to price and income changes?** The answer is elasticity — a unit-free measure of responsiveness that transforms the qualitative picture of Week 3 into quantitative predictions. We also measure the consumer’s *benefit* from a market using consumer surplus, and show how individual demand curves aggregate into a market demand curve.

Questions This Chapter Will Answer

- Why does a 10% price rise barely dent cigarette sales but devastate airline ticket sales?
- When should a firm raise its price to increase revenue — and when should it cut?
- How do we measure how much consumers benefit from a market in dollar terms?
- If your income rises 10%, do you spend proportionally more on holidays than on groceries?
- When Uber raises surge prices, does that make existing riders better or worse off?
- How do we combine two consumers' demand curves into a single market demand curve?

Chapter Road Map. Chapter 4 asks: *how much* does consumer behaviour change when prices or income change? Elasticity is our measuring tool. We define own-price elasticity, income elasticity, and cross-price elasticity, linking each to real pricing and business decisions. We then connect elasticity to total revenue — the key pricing rule: cut price if demand is elastic, raise it if inelastic. Finally, we measure the dollar value of consumer benefit using consumer surplus and show what happens to it when prices change, before closing with how individual demand curves aggregate into a market demand curve.

Learning Objectives and Topics

Each objective states what you will be able to *do*. Sections are listed beneath.

LO1 Explain why elasticity is needed and compute the point and arc price elasticity of demand.
 §4.1 Motivation §4.2 Price Elasticity

LO2 Identify the elastic, unit-elastic, and inelastic regions of a linear demand curve and explain why elasticity varies.
 §4.3 Elasticity Along Linear Demand

LO3 Classify a good by its elasticity value and explain the key determinants of price elasticity.
 §4.4 Interpreting Values §4.5 Determinants

LO4 Predict the effect of a price change on total

revenue using the elasticity-revenue relationship.
 §4.6 Elasticity and Revenue

LO5 Compute income and cross-price elasticities and use them to classify goods.
 §4.7 Income Elasticity §4.8 Cross-Price Elasticity

LO6 Define, calculate, and interpret consumer surplus; compute the change in CS when price changes.
 §4.9 Consumer Surplus

LO7 Derive a market demand curve by horizontally summing individual demand curves.
 §4.10 Market Demand §4.11 Demand Shifts

Notation Used in This Chapter.

Symbol	Meaning	Symbol	Meaning
e_d	price elasticity of demand	WTP	willingness to pay
e_I	income elasticity of demand	CS	consumer surplus
e_{xy}	cross-price elasticity (X w.r.t. P_y)	ΔCS	change in consumer surplus
TR	total revenue = $P \times Q$	Q^d	quantity demanded
$\% \Delta Q$	percentage change in quantity	Q_M	market quantity demanded
$\% \Delta P$	percentage change in price	$\bar{P}$	midpoint price (arc elasticity)

Quick Recap from Week 3. You derived the individual demand curve by tracing how the optimal bundle changes as P_x varies. A price decrease pivots the budget line outward, the consumer re-optimises, and the new (X^*, P_x) pair gives one point on the demand curve. Repeating for all prices traces the full curve. Income changes shift the entire curve: rightward for normal goods, leftward for inferior goods. The Slutsky decomposition separates a price change into a substitution effect (always toward the cheaper good) and an income effect (direction depends on good type). *This week we ask: by **how much** does quantity respond to price and income changes?*

4.1. Motivation: The Need for a Standardised Measure

If the price of bread rises by \$0.50 and bread sales fall by 1,000 loaves, while the price of semiconductors rises by \$50 and semiconductor sales fall by 500 chips — which market's consumers are more responsive to price? Raw numbers cannot answer this. We need a unit-free measure.

Up to this point, our demand analysis has relied on comparing raw quantity changes. This creates a practical problem: how do we compare responsiveness across different markets when goods are measured in different units?

What we need is a **standardised, unit-free measure** of responsiveness. That measure is **elasticity** — expressing everything as percentage changes, which are dimensionless and allow clean comparisons across any two goods.

This week we study three types of elasticity and introduce consumer surplus:

- **Price elasticity of demand** — how responsive Q^d is to own-price changes.
- **Income elasticity** — how responsive Q^d is to income changes.
- **Cross-price elasticity** — how responsive Q^d for one good is to the price of another.
- **Consumer surplus** — the dollar measure of consumers' benefit from a market.

Mini Case Study: A Coffee Shop Adjusts Prices. Sunrise Café raises its latte price from \$4.50 to \$5.00 (an 11% increase). Sales fall from 120 to 108 cups/day (a 10% drop). Revenue before:

$\$4.50 \times 120 = \540 . Revenue after: $\$5.00 \times 108 = \540 . Revenue barely changed. The price elasticity of demand is approximately $-10\%/11\% \approx -0.9$ — just inside the inelastic range. Had the owner known this, she would have predicted that a price rise near unit elasticity leaves revenue roughly unchanged. Elasticity turns this intuition into a precise business decision.

Check Your Understanding

1. Why is comparing raw quantity changes across markets misleading? Give a concrete example.
2. Percentage changes are “dimensionless.” What does this mean and why does it matter for cross-market comparison?
3. In the Sunrise Café example, compute the percentage change in price and percentage change in quantity. What is the implied elasticity?

We now know why we need elasticity. The next section defines the price elasticity of demand precisely and shows how to compute it.

4.2. Price Elasticity of Demand

A 10% price rise causes sales to fall by 20% in one market and by 2% in another. How do we capture this difference in a single number — and what does that number tell a manager about pricing strategy?

4.2.1. Intuition First

Intuition Before the Formula. Price elasticity answers: *if I raise my price by 1%, how many percentage points will my sales fall?* If sales fall by more than 1% (elastic), consumers are very sensitive — raising price hurts revenue. If sales fall by less than 1% (inelastic), consumers are relatively unresponsive — raising price increases revenue. This single number summarises everything a manager needs to know about the demand side when making pricing decisions.

4.2.2. Definition

Price Elasticity of Demand (e_d): The percentage change in quantity demanded in response to a one percent change in own price.

$$e_d = \frac{\% \Delta Q_x}{\% \Delta P_x} = \frac{\Delta Q_x / Q_x}{\Delta P_x / P_x} = \frac{\Delta Q_x}{\Delta P_x} \cdot \frac{P_x}{Q_x} \quad (4.1)$$

In words: Elasticity = (slope of demand, inverted) $\times$ (price-to-quantity ratio at the point of interest).

4.2.3. The Sign of Elasticity

e_d is **negative** for any downward-sloping demand curve. This follows from the law of demand: price and quantity move in opposite directions.

$e_d < 0$ always for downward-sloping demand. We compare $|e_d|$ to 1.

Interpretation Example: Suppose $e_d = -2$.

- A 10% *increase* in price leads to a 20% *decrease* in quantity demanded.
- A 5% *decrease* in price leads to a 10% *increase* in quantity demanded.

The negative sign confirms price and quantity move in opposite directions.

4.2.4. The Arc Elasticity Formula

When computing elasticity between two discrete points (rather than at a single point), we use the **arc (midpoint) formula** to avoid getting different answers depending on which direction we measure:

$$e_d^{\text{arc}} = \frac{(Q_2 - Q_1)/\bar{Q}}{(P_2 - P_1)/\bar{P}} \quad \text{where } \bar{Q} = \frac{Q_1 + Q_2}{2}, \bar{P} = \frac{P_1 + P_2}{2} \quad (4.2)$$

In words: Arc elasticity uses the midpoint of the price range and the midpoint of the quantity range as the reference, so the answer is the same whether price rises or falls.

Common Mistake: Arc vs. Point Elasticity. The point elasticity formula (Equation 4.1) uses a specific (P, Q) pair and the slope. The arc formula (Equation 4.2) uses midpoints between two observations. Using the wrong formula — or forgetting midpoints in the arc formula — is one of the most frequent errors on elasticity questions.

Worked Example: Computing Arc Price Elasticity (Midpoint Method).

Setup: A coffee shop lowers its price from \$4 to \$3. Daily cups sold rise from 80 to 120.

Using the midpoint formula:

$$e_d^{\text{arc}} = \frac{(Q_2 - Q_1)/\bar{Q}}{(P_2 - P_1)/\bar{P}} = \frac{(120 - 80)/100}{(3 - 4)/3.5} = \frac{0.40}{-0.286} \approx -1.40$$

where $\bar{Q} = (80 + 120)/2 = 100$ and $\bar{P} = (4 + 3)/2 = 3.50$.

Interpretation: $|e_d| = 1.40 > 1 \Rightarrow$ demand is *elastic*. A 1% price cut raises quantity by 1.4%.

Revenue check: TR before = $\$4 \times 80 = \320 ; TR after = $\$3 \times 120 = \360 . Revenue *rose* — consistent with elastic demand responding to a price cut. ✓

Two Components of Elasticity. The formula $e_d = (\Delta Q/\Delta P) \cdot (P/Q)$ has two independent parts. The first, $\Delta Q/\Delta P$, is determined entirely by the slope of the demand curve and is constant for a linear demand. The second, P/Q , depends on *where* you are on the curve. This is why elasticity varies along a linear demand even though the slope does not.

Application: Real-World Price Elasticity Estimates. Empirical economists have estimated price elasticities for hundreds of goods. Some well-established estimates from the literature:

Good	Estimated $ e_d $	Classification
Cigarettes (short run)	$\approx 0.3\text{--}0.5$	Inelastic
Petrol / gasoline	$\approx 0.2\text{--}0.4$	Highly inelastic
Electricity (residential)	$\approx 0.5\text{--}0.7$	Inelastic
Restaurant meals	$\approx 0.7\text{--}1.0$	Near unit elastic
Airline tickets (leisure)	$\approx 1.3\text{--}2.0$	Elastic
Foreign holidays	$\approx 1.5\text{--}2.5$	Elastic

Note: These are approximate estimates from empirical studies (Andreyeva et al. 2010; Brons et al. 2008; Espey 1998). Elasticities vary by country, time period, and how the market is defined. The pattern is clear: necessities with few substitutes (petrol, cigarettes) are inelastic; discretionary goods with many alternatives (holidays, airline tickets) are elastic.

Check Your Understanding

1. Price falls from \$10 to \$8; quantity rises from 40 to 50. Compute the arc elasticity using the midpoint formula. Is demand elastic or inelastic?
2. For a linear demand $Q = 80 - 4P$: compute the point elasticity at $P = 5$ and at $P = 15$.
3. A coffee chain's demand has $e_d = -0.8$. If it raises price by 5%, by what percentage will sales change? Will revenue rise or fall?

We now know how to compute elasticity at a single point. The next section shows how elasticity varies systematically along a linear demand curve, revealing why the same demand curve can be elastic at one price and inelastic at another.

4.3. Elasticity Along a Linear Demand Curve

A demand curve has a constant slope. Yet elasticity changes at every point along it. How can that be — and what does it imply about where on the demand curve a firm should operate?

4.3.1. The Key Insight

For a linear demand curve $Q = a - bP$ (with constant slope $-b$):

$$e_d = \underbrace{\frac{\Delta Q}{\Delta P}}_{\text{constant} = -b} \cdot \underbrace{\frac{P}{Q}}_{\text{varies}}$$

The slope term $(-b)$ is fixed. Elasticity varies only because P/Q changes at every point. Since P is high and Q is low at the top of the curve, P/Q is large and demand is elastic. Since P is low and Q is high at the bottom, P/Q is small and demand is inelastic.

The Three Regions of a Linear Demand Curve:

- **Upper half** (high P , low Q): $|e_d| > 1$ — **elastic**.
- **Midpoint** ($P = a/2b$, $Q = a/2$): $|e_d| = 1$ — **unit elastic**.
- **Lower half** (low P , high Q): $|e_d| < 1$ — **inelastic**.

Elasticity varies along a linear demand even though the slope does not.

Using our benchmark demand $Q = 100 - 10P$ (intercepts: $P = 10$, $Q = 100$; midpoint: $P = 5$, $Q = 50$):

In plain English:
Upper half of demand curve = elastic. Lower half = inelastic. Midpoint = unit elastic.

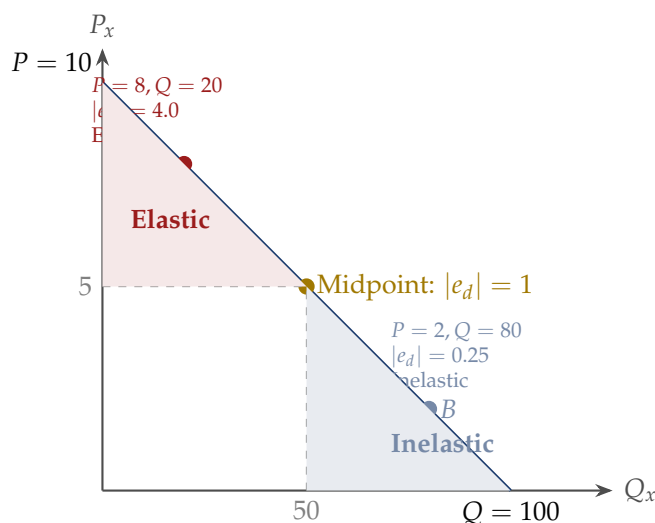


Figure 4.1. Elasticity varies along linear demand $Q = 100 - 10P$. *What to look for:* Both shaded regions have the same slope, but point $A = (P = 8, Q = 20)$ is in the elastic region ($|e_d| = 4.0$) while point $B = (P = 2, Q = 80)$ is in the inelastic region ($|e_d| = 0.25$). The midpoint ($P = 5, Q = 50$) is exactly unit elastic. All values are analytically verified: $e_d = -10 \times (P/Q)$.

4.3.2. Worked Example: Elasticity at Multiple Points

Example: Demand is $Q = 100 - 10P$. Compute e_d at $P = 2$, $P = 5$, and $P = 8$.

$\Delta Q / \Delta P = -10$ (constant slope term).

P	$Q = 100 - 10P$	$\Delta Q / \Delta P$	P/Q	e_d
2	80	-10	$2/80 = 0.025$	-0.25 (inelastic)
5	50	-10	$5/50 = 0.10$	-1.00 (unit elastic)
8	20	-10	$8/20 = 0.40$	-4.00 (elastic)

Unit-elastic point: Set $e_d = -1$: $-10 \cdot P / (100 - 10P) = -1 \Rightarrow 10P = 100 - 10P \Rightarrow P = 5$, $Q = 50$. This is the midpoint: P -intercept is 10, midpoint is $P = 5$. ✓

In words: At the midpoint of the demand curve, a 1% price rise causes exactly a 1% drop in sales — and total revenue is unchanged.

Try It. Demand is $Q = 60 - 3P$. (a) Compute e_d at $P = 5$ and $P = 15$. (b) At what price is demand unit elastic? (c) Verify using $P = a/(2b)$ where $Q = a - bP$.

Economic Insight. For any linear demand $Q = a - bP$, the revenue-maximising price is always at the midpoint $P^* = a/(2b)$, where $|e_d| = 1$. Above this price, demand is elastic and a price cut raises revenue. Below it, demand is inelastic and a price rise raises revenue. This is why profit-maximising firms with market power often aim to operate in the elastic region — they have not yet exhausted the revenue gains from cutting price.

Check Your Understanding

1. For $Q = 100 - 10P$: at what price is $|e_d| = 2$? At what price is $|e_d| = 0.5$?
2. A student says: “A steeper demand curve always has lower elasticity.” True or false? Explain using the formula.
3. Two demand curves have the same slope but different intercepts. Will they have the same elasticity at the same price? Why or why not?

We can now compute elasticity and identify regions. The next section gives names to the key values and explains the two special extreme cases (perfectly elastic and perfectly inelastic) that are important for later in the course.

4.4. Interpreting Elasticity Values

When we say demand is “elastic,” what exactly does that mean in practice — and what do the two extreme cases (perfectly inelastic and perfectly elastic) look like on a diagram?

Classification of Price Elasticity:

- **Elastic:** $|e_d| > 1$. Quantity changes more than proportionally to price.
- **Inelastic:** $|e_d| < 1$. Quantity changes less than proportionally to price.
- **Unit elastic:** $|e_d| = 1$. Quantity changes exactly proportionally to price.
- **Perfectly inelastic:** $|e_d| = 0$. Vertical demand curve — quantity does not respond to price at all.
- **Perfectly elastic:** $|e_d| \rightarrow \infty$. Horizontal demand curve — consumers buy any amount at one price but nothing above it.

Check Your Understanding

1. Match each good to its likely elasticity classification: (a) insulin for Type-1 diabetics; (b) a specific brand of bottled water; (c) wheat (the commodity).
2. Why is perfectly elastic demand relevant to Week 6 (perfect competition)? What does it mean for a competitive firm’s pricing power?
3. If $|e_d| = 0.6$ and the firm raises price by 8%, by how much does quantity fall?

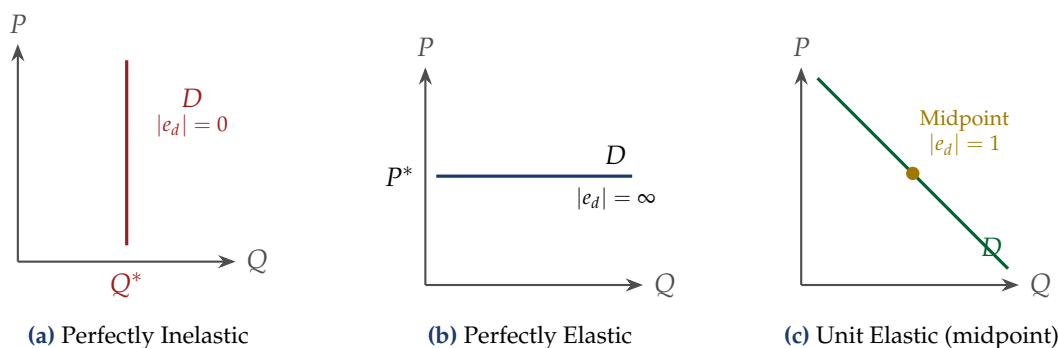


Figure 4.2. Three special elasticity cases. What to look for: (a) Vertical demand — insulin for a diabetic in the short run has no substitutes; quantity demanded is Q^* regardless of price. (b) Horizontal demand — a price-taker in a perfectly competitive market; any price above P^* drives sales to zero. (c) The midpoint of a linear demand is exactly unit elastic.

Does revenue rise or fall?

We can now classify goods by elasticity. But what determines whether demand is elastic or inelastic in the first place? The next section identifies the key factors.

4.5. Determinants of Price Elasticity of Demand

Why does demand for petrol barely respond to a 20% price rise, while demand for a particular airline route collapses? What economic forces determine whether consumers are responsive or not?

Two key factors explain most variation in elasticity across goods and markets.

1. Number of close substitutes. If a good has many close substitutes, consumers easily switch when price rises — demand is **elastic**. If substitutes are scarce, consumers keep buying despite higher prices — demand is **inelastic**.

Example: Demand for “food” is inelastic (everyone needs to eat). Demand for “Italian restaurant meals” is elastic (there are many alternatives). The narrower the market definition, the more substitutes exist and the more elastic demand tends to be.

2. Time horizon. The **longer the time horizon**, the **more elastic** demand becomes. Over time, consumers can find substitutes, change habits, or buy substitute goods (e.g. switching to a fuel-efficient car after sustained high petrol prices).

Classic example: Short-run demand for petrol is highly inelastic (people still drive to work). Long-run demand is more elastic as consumers buy smaller cars, carpool, or relocate.

Factor	More elastic	More inelastic
Substitutes	Many close substitutes	Few substitutes
Time horizon	Long run	Short run
Market breadth	Narrow definition	Broad definition

Common Mistake: Confusing the market with the brand. “Demand for food is elastic because food prices fluctuate” is wrong. Demand for *food in general* is highly inelastic — everyone needs to eat. It is demand for a *specific product or brand* that can be very elastic, because substitutes abound. Always ask: how narrowly is the market defined?

Why This Matters: Elasticity in Pricing Strategy. Firms use knowledge of price elasticity every time they set prices. A pharmaceutical company with a patented drug and no substitutes faces inelastic demand — it can raise price with minimal sales loss. A grocery store competing with several nearby chains faces elastic demand and cannot raise prices without losing customers to rivals. Knowing your product’s elasticity is step one in any pricing strategy.

Check Your Understanding

1. Rank by expected $|e_d|$ from lowest to highest: (a) salt; (b) a specific brand of salt; (c) restaurant meals; (d) insulin. Justify each.
2. A government considers taxing petrol to reduce carbon emissions. Why would the short-run reduction in petrol consumption be smaller than the long-run reduction?
3. A firm discovers its demand elasticity is -0.4 . Should it raise or lower its price to increase revenue? By contrast, what should a firm with elasticity -1.8 do?

We now understand what determines elasticity. The single most important application is the relationship between elasticity and total revenue. This is the key insight that turns elasticity from a theoretical concept into a practical pricing tool.

4.6. Elasticity and Total Revenue

A firm is deciding whether to raise or cut its price. It knows demand will respond — but will the revenue go up or down? The answer depends entirely on one number: the price elasticity of demand.

4.6.1. Motivation

Total revenue (from the firm’s perspective) equals total expenditure (from the consumer’s):

$$TR = P \times Q \quad (4.3)$$

In words: Revenue = price per unit $\times$ units sold. A price rise increases each unit’s contribution but reduces how many units are sold. The net effect depends on which force dominates.

When price rises, two opposing forces act on TR :

- **Price effect:** Each unit sold now brings in more revenue ($\uparrow$).
- **Quantity effect:** Fewer units are sold ($\downarrow$).

Which effect dominates depends on whether demand is elastic or inelastic.

The Elasticity–Revenue Summary Table

	Elastic ($ e_d > 1$)	Unit Elastic ($ e_d = 1$)	Inelastic ($ e_d < 1$)
Price $\uparrow$	TR $\downarrow$	TR unchanged	TR $\uparrow$
Price $\downarrow$	TR $\uparrow$	TR unchanged	TR $\downarrow$

Managerial rule: Raise price if demand is inelastic. Cut price if demand is elastic. At unit elastic, any price change leaves TR unchanged.

4.6.2. When Demand is Elastic ($|e_d| > 1$): Quantity Effect Dominates

When $|e_d| > 1$, a 1% price rise causes more than a 1% drop in quantity. The quantity loss outweighs the price gain:

TR $\downarrow$ when price rises and demand is elastic.

Conversely, a price cut raises TR: the quantity gain exceeds the revenue lost per unit.

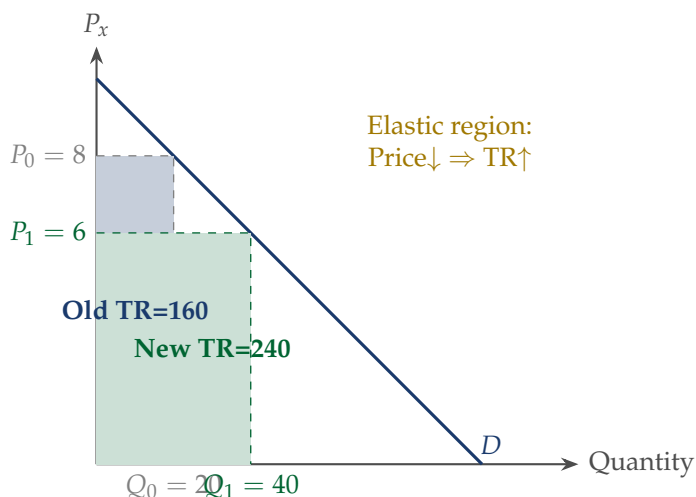


Figure 4.3. Elastic demand: price cut raises revenue. What to look for: P falls from 8 to 6 (25% drop) while Q rises from 20 to 40 (100% rise) — the quantity gain far exceeds the price loss. TR rises from \$160 to \$240. Verified: $e_d = -10 \times 8/20 = -4.0$ (elastic). ✓

4.6.3. When Demand is Inelastic ($|e_d| < 1$): Price Effect Dominates

When $|e_d| < 1$, a 1% price rise causes less than a 1% drop in quantity. The price gain outweighs the quantity loss:

TR $\uparrow$ when price rises and demand is inelastic.

The Elasticity–Revenue Rule. When you raise price, two forces pull in opposite directions: the higher price per unit pushes revenue up, but fewer units sold pushes it down. If $|e_d| > 1$, the quantity loss wins and revenue falls. If $|e_d| < 1$, the price gain wins and revenue rises.

Only at $|e_d| = 1$ do the two forces exactly offset, leaving revenue unchanged. This tipping point is exactly the midpoint of a linear demand curve.

Reverse Problem: Back out elasticity from revenue data.

A firm observes that raising price by 10% reduced total revenue by 12%. What does this tell us about the elasticity of demand?

Step 1: Revenue fell when price rose $\Rightarrow$ demand is **elastic** ($|e_d| > 1$).

Step 2: $TR = P \times Q$. With $\Delta P/P = +10\%$ and $\Delta TR/TR = -12\%$:

$$\% \Delta TR \approx \% \Delta P + \% \Delta Q \Rightarrow -12\% \approx +10\% + \% \Delta Q \Rightarrow \% \Delta Q \approx -22\%$$

Step 3: $|e_d| = |\% \Delta Q / \% \Delta P| \approx 22/10 = 2.2$.

Economic Insight. This reverse-engineering approach is how real firms estimate demand elasticity from market data. Rather than running a controlled experiment, they observe revenue changes after price changes and infer the elasticity. The key: if TR and P move in opposite directions, demand is elastic; if they move together, demand is inelastic.

Application: Old Navy — Should We Raise or Lower Prices? Suppose Old Navy consults you on pricing strategy. Clothing has many substitutes (other brands, second-hand markets, fast fashion) — suggesting demand is relatively *elastic* ($|e_d| > 1$). If elastic, *raising* prices would *reduce* total revenue. The better strategy is to cut prices or run promotions — the increase in volume more than compensates for the lower margin per unit. Alternatively, Old Navy could invest in brand differentiation to make demand more inelastic before raising prices. This is exactly the strategic logic behind luxury brand positioning.

Check Your Understanding

1. For $Q = 100 - 10P$: compute TR at $P = 3$, $P = 5$, and $P = 7$. At which price is TR maximised? What is the elasticity at that price?
2. A firm raises price by 15% and finds revenue rises by 6%. Is demand elastic, inelastic, or unit elastic? Estimate $|e_d|$ approximately.
3. A government taxes cigarettes (inelastic demand) and restaurant meals (near unit elastic). In which case will tax revenue be larger relative to the quantity reduction? Explain using the elasticity-revenue relationship.

Price elasticity measures responsiveness to own-price changes. But demand also responds to income and to the prices of other goods. The next two sections introduce those two other elasticities.

4.7. Income Elasticity of Demand

When household incomes double, does demand for restaurant meals double too? What about demand for instant noodles — or bus travel? The income elasticity of demand answers these questions and classifies goods into luxuries, necessities, and inferior goods.

4.7.1. Intuition First

Intuition Before the Formula. Income elasticity answers: if consumer incomes rise by 1%, by what percentage does demand for this good change? If demand rises by more than 1% (luxury), the good is disproportionately desired by richer consumers. If demand rises by less than 1% (necessity), everyone needs it but the rich don't buy much more of it. If demand falls (inferior good), richer consumers trade up to better alternatives.

4.7.2. Definition and Classification

Income Elasticity of Demand (e_I):

$$e_I = \frac{\% \Delta Q_x}{\% \Delta I} = \frac{\Delta Q_x}{\Delta I} \cdot \frac{I}{Q_x} \quad (4.4)$$

In words: Income elasticity = percentage change in quantity demanded per one percentage point rise in income.

- **Normal goods:** $e_I > 0$ — demand rises with income.
- **Inferior goods:** $e_I < 0$ — demand falls as income rises.
- **Luxury goods:** $e_I > 1$ — demand rises faster than income (a subcategory of normal goods).
- **Necessities:** $0 < e_I < 1$ — demand rises, but more slowly than income.

Example: Demand for restaurant meals is $Q = 50 + 0.4I - 5P$. At $I = 3,000$ and $P = 20$: $Q = 1,150$. Income elasticity:

$$e_I = \frac{\partial Q}{\partial I} \cdot \frac{I}{Q} = 0.4 \cdot \frac{3000}{1150} \approx 1.04$$

Since $e_I \approx 1.04 > 1$, restaurant meals are a **luxury good** for this consumer at these parameters.

*Luxury: $e_I > 1$.
Necessity:
 $0 < e_I < 1$.
Inferior: $e_I < 0$.*

Application: Income Elasticity and the Business Cycle.

Income elasticity determines how exposed a firm is to macroeconomic fluctuations:

Category	e_I range	Business cycle sensitivity
Luxury goods	> 1	High — booms great, recessions devastating
Necessities	$0-1$	Low — stable through cycles
Inferior goods	< 0	Counter-cyclical — recession-proof

Examples: Fine dining, luxury cars, and foreign holidays ($e_I > 1$) collapsed during the 2008 recession. Supermarkets and discount retailers (e_I near 0 or negative) performed well. Spam canned meat and instant noodles saw demand *rise* during downturns. Investment analysts use income elasticities to predict which sectors will outperform during recessions versus expansions.

Going deeper: Engel curves plot quantity demanded against income, holding prices fixed. For a normal

good the Engel curve slopes upward; for an inferior good it eventually bends back. The income elasticity at any point equals the slope of the Engel curve times I/Q — a direct graphical analogue of the formula above.

Check Your Understanding

1. At $I = \$4,000$, demand for streaming subscriptions is $Q = 6$ (from $Q = 2 + 0.001I$). Compute e_I . Is this a luxury, necessity, or inferior good?
2. A consumer buys 10 bus rides per week at $I = \$500$ and 6 rides at $I = \$800$. Compute e_I approximately. Classify the good.
3. A portfolio manager wants to invest in recession-resistant consumer goods. Should she favour goods with $e_I > 1$ or $e_I < 1$? Explain.

Income elasticity classifies goods by how income-responsive they are. Cross-price elasticity classifies goods by how they relate to each other — substitutes or complements.

4.8. Cross-Price Elasticity of Demand

When Netflix raises its subscription price, does demand for Disney+ rise or fall? When petrol prices spike, does demand for electric vehicles rise? The cross-price elasticity of demand answers these market-linkage questions.

4.8.1. Intuition First

Intuition Before the Formula. Cross-price elasticity answers: if the price of good Y rises by 1%, by what percentage does demand for good X change? If demand for X rises (substitutes), consumers are switching from Y to X because Y got more expensive. If demand for X falls (complements), X and Y are used together, so when Y becomes expensive and people buy less Y , they also buy less X .

4.8.2. Definition

Cross-Price Elasticity of Demand (e_{xy}):

$$e_{xy} = \frac{\% \Delta Q_x}{\% \Delta P_y} = \frac{\Delta Q_x}{\Delta P_y} \cdot \frac{P_y}{Q_x} \quad (4.5)$$

In words: Cross-price elasticity = percentage change in demand for X per one percentage point rise in the price of Y .

The **sign** carries all the information:

- $e_{xy} > 0$: **Substitutes** — $P_y \uparrow$ causes consumers to switch toward X .
- $e_{xy} < 0$: **Complements** — $P_y \uparrow$ reduces demand for Y and hence also for X .
- $e_{xy} \approx 0$: **Unrelated goods**.

Relationship	Sign of e_{xy}	Example
Substitutes	+	Butter & margarine; Netflix & Disney+
Complements	−	Cars & petrol; printers & ink
Unrelated	≈ 0	Books & gasoline

Common Mistake: Sign error on complements. Students sometimes expect printers and ink to have a positive cross-price elasticity because their prices are often promoted together. They do not. When printer prices rise, fewer printers are sold, so demand for ink falls. $e_{xy} < 0$ confirms they are complements. **Always trace through the demand logic: does higher P_y cause consumers to buy more or less of X ?**

Business application: Cross-price elasticity is heavily used in antitrust economics. When two firms compete and one raises prices, if demand for the other rises significantly (high positive e_{xy}), they are in the same market. Regulators use this to define market boundaries in merger reviews.

Check Your Understanding

1. Netflix raises its price by 10%; demand for Disney+ rises by 8%. Compute e_{xy} . Substitutes or complements?
2. Petrol price rises by 20%; demand for electric vehicles rises by 15%. What does this imply about the relationship between petrol and petrol-powered cars? Between petrol and electric vehicles?
3. A regulator argues that Product A and Product B are in the same market because their cross-price elasticity is +2.1. Is this evidence sufficient? What would a negative cross-price elasticity tell you?

We have now covered all three elasticities. We turn to one of the most important welfare concepts in economics: consumer surplus, which uses the demand curve to put a dollar value on how much consumers benefit from a market.

4.9. Consumer Surplus

You would have paid \$15 for a book that cost you \$10 in the market. You got \$5 worth of surplus. How do we aggregate this across all buyers and all units to measure the total benefit consumers receive from a market?

4.9.1. Willingness to Pay

The demand curve encodes a willingness-to-pay (WTP) schedule: the height of the demand curve at any quantity Q is the maximum price a consumer would pay for the Q -th unit.

Declining WTP reflects diminishing marginal utility: as you consume more of a good, each additional unit is worth less to you.

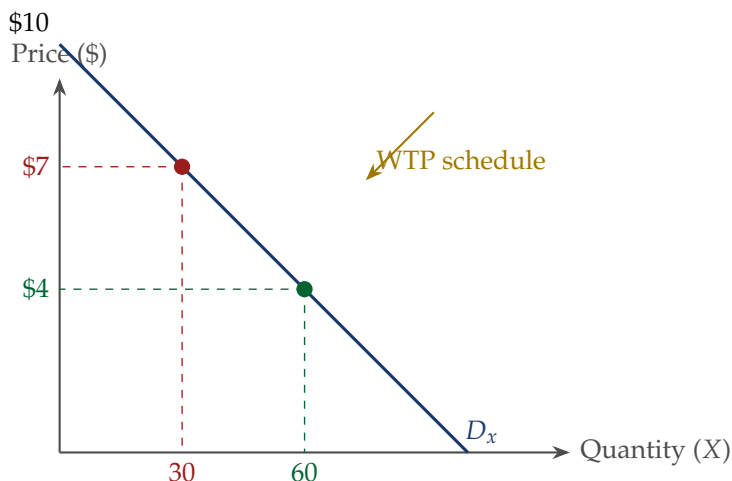


Figure 4.4. Demand curve as willingness-to-pay schedule. At 30 units, WTP = \$7; at 60 units, WTP = \$4. The choke price (maximum WTP) is \$10. For demand $P = 10 - Q/10$, WTP falls as quantity rises — reflecting diminishing marginal utility.

4.9.2. Definition and Formula

Consumer Surplus (CS): The aggregate difference between consumers' willingness to pay and the price they actually pay.

$$CS = \text{Total WTP} - \text{Total Amount Paid} = \int_0^{Q^*} D(Q) dQ - P^* \cdot Q^* \quad (4.6)$$

For a **linear demand** $P = a - bQ$ at market price P^* :

$$CS = \frac{1}{2}(a - P^*) \cdot Q^* \quad (4.7)$$

In words: Consumer surplus is the area of the triangle above the market price and below the demand curve — the aggregate gain to all buyers from paying less than their maximum willingness to pay.

$CS = \text{triangle area}$
 $= \frac{1}{2} \times \text{base} \times \text{height}.$

4.9.3. Graphical Representation and Solved Example

Solved Example: Computing Consumer Surplus.

Demand: $P = 10 - Q/10$, so $Q = 100 - 10P$. Market price $P^* = \$5$.

Step 1: Find Q^* : $Q^* = 100 - 10(5) = 50$ units.

Step 2: Identify triangle dimensions:

- **Height** = $P_{\max} - P^* = 10 - 5 = \5 (gap between choke price and market price).
- **Base** = $Q^* = 50$ units.

Step 3: Apply formula (Equation 4.7):

$$CS = \frac{1}{2} \times 50 \times 5 = \$125$$

In words: At a price of \$5, consumers collectively receive \$125 in surplus — they would have collectively been willing to pay \$125 more than they actually did.

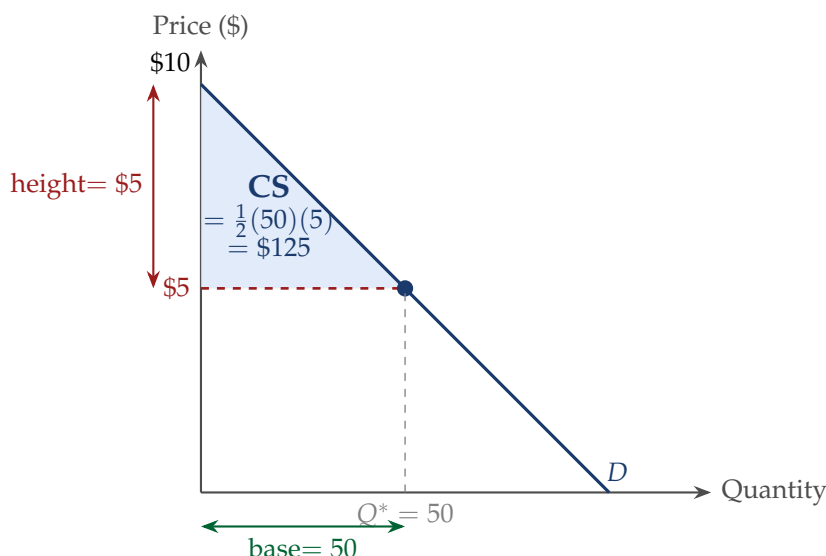


Figure 4.5. Consumer surplus as a triangle. *What to look for:* Demand $P = 10 - Q/10$, market price \$5, $Q^* = 50$. The shaded triangle is $CS = \frac{1}{2} \times 50 \times (10 - 5) = \125 . The height is the gap between the choke price (\$10) and the market price (\$5). All coordinates verified analytically.

Worked Example: Consumer Surplus and the Effect of a Price Rise.

Setup: Market demand is $Q^D = 100 - 2P$. Current price is $P^* = \$20$.

Step 1: Find equilibrium quantity: $Q^* = 100 - 2(20) = 60$.

Step 2: Find the demand intercept (reservation price): set $Q = 0 \Rightarrow P_{\max} = 50$.

Step 3: $CS = \frac{1}{2}(P_{\max} - P^*) \cdot Q^* = \frac{1}{2}(50 - 20)(60) = \frac{1}{2}(30)(60) = \900 .

Interpretation: Consumers collectively enjoy \$900 of surplus — the dollar value of benefit they receive above what they actually pay.

Now suppose price rises to $P = \$30$:

New $Q^* = 100 - 2(30) = 40$. New $CS = \frac{1}{2}(50 - 30)(40) = \400 .

Loss in CS = $\$900 - \$400 = \$500$ (composed of a rectangle transferred to producers and a DWL triangle of buyers who exit).

Elasticity check: At the original point ($P = 20, Q = 60$): $e_d = -2 \times (20/60) = -0.67$ — inelastic, so raising price increases revenue. The loss in CS is large nonetheless, because many consumers were buying at \$20 and must now pay more or drop out.

4.9.4. Effect of Price Changes on Consumer Surplus

When price falls from P_1 to $P_2 < P_1$, consumer surplus increases by a trapezoid:

$$\Delta CS = \underbrace{(P_1 - P_2) \cdot Q_1}_{\text{Rectangle}} + \underbrace{\frac{1}{2}(P_1 - P_2)(Q_2 - Q_1)}_{\text{Triangle}} \quad (4.8)$$

In words: The gain in CS has two parts: existing buyers pay less (rectangle gain), and new buyers enter who previously couldn't afford the good (triangle gain).

Solved Example: Change in CS when price rises.

Demand: $Q = 100 - 10P$. Price rises from $P_1 = \$5$ to $P_2 = \$7$.

- $Q_1 = 100 - 50 = 50$; $Q_2 = 100 - 70 = 30$.
- CS at $P_1 = 5$: $\frac{1}{2}(50)(10 - 5) = \125 .
- CS at $P_2 = 7$: $\frac{1}{2}(30)(10 - 7) = \45 .
- **Loss in CS** = $125 - 45 = \$80$.

Decomposed (Equation 4.8):

- Rectangle: $(7 - 5) \times 30 = \$60$ (existing buyers pay more).
- Triangle: $\frac{1}{2}(7 - 5)(50 - 30) = \20 (buyers who exit the market).
- Total loss: $60 + 20 = \$80$. ✓

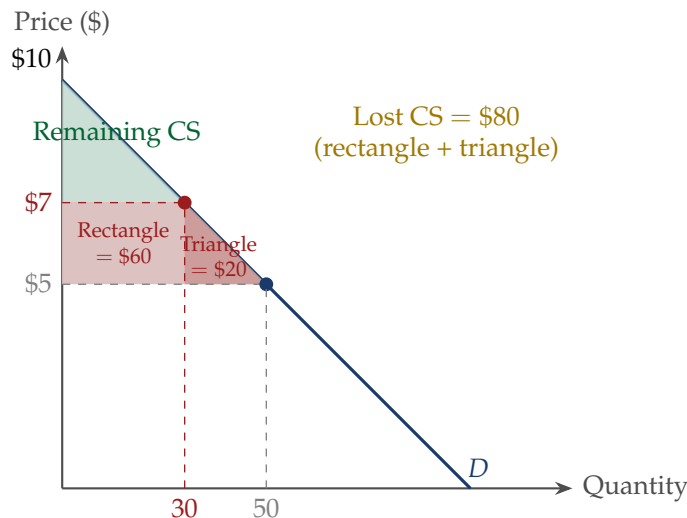


Figure 4.6. Loss in consumer surplus when price rises from \$5 to \$7. What to look for: The red shaded area (\$80) is lost CS. It splits into a rectangle (\$60 — existing buyers paying more) and a triangle (\$20 — buyers who exit the market). All values verified: $CS_1 = \$125$, $CS_2 = \$45$, loss = \$80. ✓

Why This Matters: Consumer Surplus and Policy. Consumer surplus is the standard tool economists use to evaluate welfare gains and losses from policy changes. When a government caps a drug price below the market price, CS rises for those who can still buy it. When a tariff raises the price of imported goods, domestic consumers lose CS. Estimating these triangles and trapezoids is at the core of cost-benefit analysis in health care, trade policy, and competition law.

Going deeper: Consumer surplus as defined here is the *Marshallian* (uncompensated) surplus. Advanced microeconomics uses two refined welfare measures: *compensating variation* (CV) — how much income would the consumer need to be compensated for a price rise to stay equally well off — and *equivalent variation* (EV)

— how much income would give the same welfare change as the price rise. For small price changes, $CS \approx CV \approx EV$, but they diverge for large changes. These concepts are central to welfare economics and cost-benefit analysis.

Check Your Understanding

1. Demand: $Q = 60 - 4P$. Market price $P^* = \$6$. Compute CS.
2. In the above, price rises to \$9. Compute the new CS. What is the loss? Decompose into rectangle and triangle.
3. Uber uses surge pricing (raising price above normal during peak demand). Using consumer surplus, explain why existing riders are worse off under surge pricing. Who might be better off?

Consumer surplus is a property of an individual demand curve. Markets, however, involve many consumers. The next section shows how to aggregate individual demand curves into a market demand curve.

4.10. Market Demand Curve

A market has thousands of consumers, each with their own demand curve. How do we combine them into a single market demand curve — and what happens when consumers have different reservation prices?

4.10.1. From Individual to Market Demand

The **market demand curve** is obtained by **horizontal aggregation**: at each price, sum the quantities demanded by all consumers.

Why Market Demand is Flatter. When price rises by \$1, each individual consumer reduces quantity by some amount. At the market level, we add up all those individual reductions. The total quantity drop is larger than any single consumer's — so the market demand curve has a flatter slope and higher elasticity than any individual's curve.

4.10.2. Aggregation with Heterogeneous Consumers: A Worked Example

When consumers have different reservation prices (choke prices), the market demand curve has a **kink** at the price where one consumer exits the market.

Solved Example: Two consumers with different demands.

Consumer 1: $Q_1 = 10 - 2P$ (exits at $P = 5$, i.e. $Q_1 = 0$ when $P \geq 5$).

Consumer 2: $Q_2 = 8 - P$ (exits at $P = 8$, i.e. $Q_2 = 0$ when $P \geq 8$).

Market demand:

Case 1 ($P > 5$): Only Consumer 2 is active. $Q_M = Q_2 = 8 - P$.

Case 2 ($P \leq 5$): Both consumers active.

$$Q_M = Q_1 + Q_2 = (10 - 2P) + (8 - P) = 18 - 3P$$

Kink check at $P = 5$:

- From Case 1: $Q_M = 8 - 5 = 3$.
- From Case 2: $Q_M = 18 - 15 = 3$. ✓ (Continuous at kink.)

P	Q_1	Q_2	Q_M
7	0	1	1
5	0	3	3
4	2	4	6
2	6	6	12

The kink appears because Consumer 1 exits at $P = 5$: for $P > 5$, the market demand curve is steeper (only one consumer); for $P \leq 5$, it becomes flatter as both consumers participate.

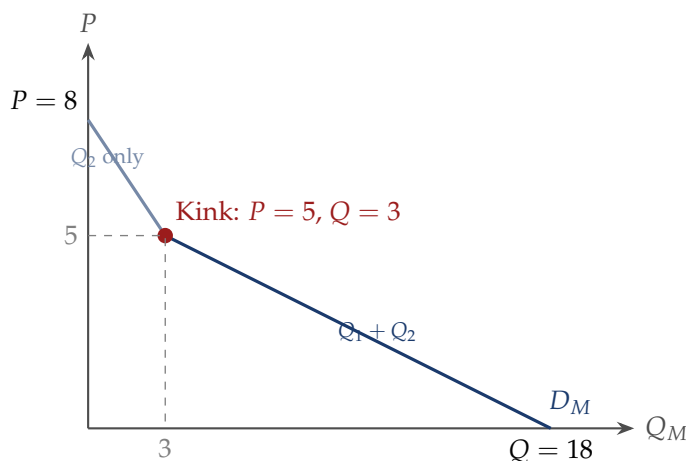


Figure 4.7. Market demand curve with a kink. *What to look for:* For $P > 5$, only Consumer 2 participates and the curve is $Q_M = 8 - P$ (steeper). For $P \leq 5$, both participate and $Q_M = 18 - 3P$ (flatter). The kink at $(Q = 3, P = 5)$ is where Consumer 1 enters the market. All values analytically exact.

Check Your Understanding

1. Consumers: $Q_1 = 12 - 3P$ (exits at $P = 4$); $Q_2 = 10 - 2P$ (exits at $P = 5$). Derive the market demand curve. At what price does the kink occur? What is the market quantity at the kink?
2. Why is the market demand curve always at least as flat as any individual demand curve? Use the aggregation logic to explain.
3. A market has 100 identical consumers each with demand $Q_i = 20 - 4P$. Write the market demand curve.

The final section shows how the market demand curve shifts when income or prices of related goods change — completing the picture of what determines market demand.

4.11. Shifts in the Market Demand Curve

A firm's demand curve was drawn holding income and other prices constant. What happens to the whole market when the economy grows, or when the price of a substitute or complement changes?

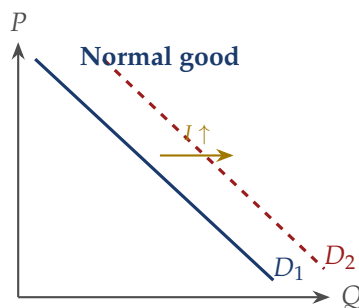
4.11.1. Changes in Income

- **Normal good:** $I \uparrow \Rightarrow$ demand shifts **right** (more demanded at every price).
- **Inferior good:** $I \uparrow \Rightarrow$ demand shifts **left** (less demanded at every price).

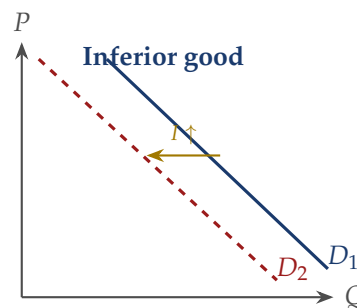
Both cases matter. If a question asks “how does market demand change when income increases?” without specifying the good type, you must address both scenarios. Presenting only one — even if correct — is an incomplete answer.

4.11.2. Changes in Prices of Related Goods

- **Substitutes:** $P_Y \uparrow \Rightarrow$ consumers switch toward $X \Rightarrow D_X$ shifts **right**.
- **Complements:** $P_Y \uparrow \Rightarrow$ demand for Y falls $\Rightarrow$ demand for X also falls $\Rightarrow D_X$ shifts **left**.



(a) Normal: demand shifts right



(b) Inferior: demand shifts left

Figure 4.8. Income increase shifts market demand. *What to look for:* In both panels, P_X is unchanged — this is not a movement along D , it is a shift of the entire curve. For normal goods (left), the shift is rightward; for inferior goods (right), leftward.

Check Your Understanding

1. State the direction of the shift in market demand for X in each case: (a) income rises, X is normal; (b) income falls, X is inferior; (c) P_Y rises, X and Y are complements; (d) P_Y falls, X and Y are substitutes.
2. Netflix raises its prices. What happens to market demand for Disney+? What about market demand for high-speed internet (a complement to streaming)?
3. During a recession (income falls), demand for budget airlines rises. What does this tell you about budget airline travel's income elasticity?

The theory is complete. Before turning to the practice questions, let us collect the key formulas, flag the most common exam mistakes, and build the revision summary.

The worked examples and applications above have shown the mechanics of all three elasticities, consumer surplus, and market demand aggregation. The practice questions that follow test all of these — both forward (given demand and prices, compute CS or elasticity) and in reverse (given observed revenue changes, recover elasticity). Attempt every question before consulting the solutions.

Where Does This Take Us?

- **Week 5** uses consumer surplus alongside *producer surplus* to evaluate total market welfare and the gains from trade. The demand curve you have now derived and the CS triangle you computed are the inputs to that analysis.
- **Price ceilings and floors** (Week 5) create deadweight loss — a reduction in the sum of consumer and producer surplus. You will compute this as a triangle, using exactly the tools from this week.
- **Tax incidence** (Week 5) splits the tax burden between consumers and producers. The relative elasticities of demand and supply determine who bears more of the burden — the more inelastic side bears more.
- **Market structure** (Week 6 onwards) uses demand elasticity to determine optimal pricing. A monopolist sets price where $|e_d| > 1$ (elastic region) — you can now verify why from the elasticity-revenue table.

Five Common Exam Mistakes — and How to Avoid Them

1. **Using point elasticity when arc elasticity is required (or vice versa).** Point elasticity uses a specific (P, Q) pair and the slope. Arc elasticity uses midpoints $\bar{P} = (P_1 + P_2)/2$ and $\bar{Q} = (Q_1 + Q_2)/2$. Always check whether the question gives you a single point or two observations.
2. **Getting the elasticity-revenue direction wrong.** Elastic demand ($|e_d| > 1$): price rise $\Rightarrow$ TR falls. Inelastic demand ($|e_d| < 1$): price rise $\Rightarrow$ TR rises. A reliable check: if in doubt, compute TR at both prices and compare directly.
3. **Sign error on cross-price elasticity for complements.** Complements have $e_{xy} < 0$, not positive. When printer prices rise, demand for ink falls. Trace through the demand logic rather than guessing from price correlations.
4. **Drawing consumer surplus below the demand curve instead of above the price line.** CS is the area *above* the market price and *below* the demand curve — the triangle between the two. It is not the area below the demand curve (which includes the expenditure rectangle too).
5. **Forgetting the kink when aggregating heterogeneous demand curves.** When consumers have different reservation prices, the market demand curve has a kink at the price where the consumer with the lower reservation price exits. For P above the lower reservation price, use only the remaining consumers' demands.

Chapter at a Glance

Your one-page revision guide — everything you need before an exam or tutorial.

Key Formulas

Price elasticity	$e_d = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$	p. 4
Arc elasticity	midpoints formula	p. 5
Income elasticity	$e_I = \frac{\Delta Q}{\Delta I} \cdot \frac{I}{Q}$	p. 13
Cross-price elast.	$e_{xy} = \frac{\Delta Q_x}{\Delta P_y} \cdot \frac{P_y}{Q_x}$	p. 14
Total revenue	$TR = P \times Q$	p. 10
Consumer surplus	$CS = \frac{1}{2}(P_{\max} - P^*)Q^*$	p. 16

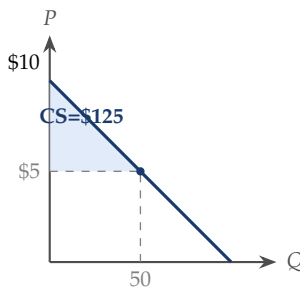
Elasticity–Revenue Table

	Elastic	Unit	Inelastic
	$ e_d > 1$	$ e_d = 1$	$ e_d < 1$
$P \uparrow$	TR↓	TR same	TR↑
$P \downarrow$	TR↑	TR same	TR↓

Good Classifications

Type	e_I range	Relationship	e_{xy}
Luxury	> 1	Substitutes	> 0
Necessity	0–1	Complements	< 0
Inferior	< 0	Unrelated	≈ 0

Consumer Surplus ($Q = 100 - 10P$, $P^* = 5$)



Linear Demand Elasticity Regions

- **Upper half** (high P , low Q): $|e_d| > 1$ — elastic
- **Midpoint**: $|e_d| = 1$ — unit elastic, max TR
- **Lower half** (low P , high Q): $|e_d| < 1$ — inelastic
- At $P = a/(2b)$ for $Q = a - bP$: $|e_d| = 1$

LOs Covered

- LO1 — motivation, point and arc elasticity ✓
- LO2 — elastic/inelastic regions ✓
- LO3 — classification and determinants ✓
- LO4 — elasticity-revenue relationship ✓
- LO5 — income and cross-price elasticity ✓
- LO6 — consumer surplus ✓
- LO7 — market demand aggregation ✓

The thread: Unit-free measure needed → define e_d → elastic/inelastic regions → determinants → elasticity-revenue rule → income & cross-price elasticity → consumer surplus (dollar value of demand) → market aggregation. Week 5 uses CS to evaluate policy welfare effects.

4.12. Key Terms

Elasticity

Unit-free measure of responsiveness; ratio of percentage changes. *Used throughout the course.*

Price Elasticity (e_d)

$\% \Delta Q / \% \Delta P_x$. Always negative for downward-sloping demand. *Used throughout the course.*

Elastic

$|e_d| > 1$. Quantity more than proportionally responsive.

Inelastic

$|e_d| < 1$. Quantity less than proportionally responsive.

Unit Elastic

$|e_d| = 1$. At midpoint of linear demand; TR is maximised here.

Perfectly Inelastic

$|e_d| = 0$. Vertical demand curve.

Perfectly Elastic

$|e_d| \rightarrow \infty$. Horizontal demand curve.

Arc Elasticity

Elasticity between two points using midpoint formula for $\bar{P}$ and $\bar{Q}$.

Total Revenue

$TR = P \times Q$. *Used throughout the course.*

Income Elasticity (e_I)

$\% \Delta Q / \% \Delta I$. Positive for normal goods;

negative for inferior goods.

Luxury Good

Normal good with $e_I > 1$; consumption rises faster than income.

Necessity

Normal good with $0 < e_I < 1$; consumption rises but less than income.

Cross-Price Elasticity (e_{xy})

$\% \Delta Q_x / \% \Delta P_y$. Positive for substitutes; negative for complements.

Willingness to Pay (WTP)

Maximum price a consumer would pay; given by demand curve height. *Used throughout the course.*

Consumer Surplus (CS)

Aggregate WTP minus amount paid; area above price and below demand curve. *Used throughout the course.*

Market Demand

Horizontal sum of all individual demand curves.

Horizontal Aggregation

Summing quantities at each price to derive market demand.

Kinked Demand

Market demand kink where a consumer exits the market; slope changes at that price.

Recommended Videos

Four videos selected for clarity and direct relevance to this chapter. Watch them in order.

1. Price Elasticity of Demand (Midpoint Method) (*Khan Academy*) ~ 7 min

<https://www.youtube.com/watch?v=s1P8XZ6Nq40>

Covers the definition, midpoint formula, and intuition for price elasticity. Directly reinforces Sections 4.2–4.3 and is the ideal starting point before working through elasticity along a linear demand curve.

2. Total Revenue and Elasticity (*Khan Academy*) ~ 10 min

<https://www.youtube.com/watch?v=Ud1tkvm9HLc>

Explains the elasticity-revenue relationship with worked examples and the revenue-rectangle diagram. Closely mirrors the analysis in Section 4.6 and is particularly valuable before attempting Level 2 revenue questions.

3. Consumer Surplus (*Khan Academy*) ~ 8 min

https://www.youtube.com/watch?v=_6kwhF6hoqQ

Visual walkthrough of consumer surplus as a triangle and the effect of price changes, with clear diagrams. Reinforces Section 4.9 and is recommended before attempting the consumer surplus questions in Level 2–3.

4. Market Demand as the Sum of Individual Demand (*Khan Academy*) ~ 9 min

<https://www.youtube.com/watch?v=dGjPguqHBXE>

Demonstrates horizontal summation of individual demand curves with a worked numerical example. Directly reinforces Section 4.10.

4.13. Practice Questions

How to use these questions. Arranged in three levels. Level 1 builds vocabulary and conceptual understanding; Level 2 requires computation and diagram work; Level 3 demands synthesis across topics. Difficulty: ★ easy ★★ intermediate ★★★ challenging. Attempt every question independently before consulting the solutions.

Level 1: Conceptual and Short-Answer

- ★ Define price elasticity of demand. Why is it typically negative? Give one example of a real good and state whether you expect its demand to be elastic or inelastic.
- ★ Explain why a linear demand curve does not have constant price elasticity even though it has a constant slope. At what point is $|e_d| = 1$?
- ★ For each factor below, state whether it makes demand more elastic or more inelastic:
 - (a) Availability of many close substitutes increases.
 - (b) Time horizon is very short.
 - (c) Good is defined very narrowly (specific brand rather than category).
 - (d) Good is a life-saving medicine with no alternative.
- ★ Complete the elasticity-revenue table and briefly justify each entry:

Scenario	Elastic or Inelastic?	Price change	TR effect
Firm A (elastic demand)	Elastic	Price ↑	?
Firm B (inelastic demand)	Inelastic	Price ↑	?
Firm A (elastic demand)	Elastic	Price ↓	?
Firm B (inelastic demand)	Inelastic	Price ↓	?

- ★ What does consumer surplus measure? Give a simple numerical example. Why is it called “notional” rather than actual cash received?
- ★ Explain horizontal aggregation. If two consumers have identical individual demand curves, what does the market demand curve look like?
- ★ What is cross-price elasticity? How does its sign identify substitutes vs. complements? Give one example of each.
- ★ **[Explain to a Friend — Level 1]** *No formulas allowed.* Your friend says: “I don’t understand why a firm with inelastic demand should raise its price. If it raises price, won’t it sell less?” Write 3–4 plain-English sentences explaining the elasticity-revenue logic.

Level 2: Intermediate Problems

- ★★ Demand: $Q = 120 - 4P$.

- (a) Compute e_d at $P = 10, P = 15, P = 25$.
 - (b) At what price is demand unit elastic?
 - (c) Graph the demand curve indicating elastic, unit-elastic, and inelastic regions.
 - (d) Compute TR at each price. Do results agree with the elasticity values?
- ★★ Two consumers: $Q_1 = 8 - 2P$ and $Q_2 = 6 - P$.
- (a) At what price does each consumer exit the market?
 - (b) Derive the market demand function (specify each price range).
 - (c) Find Q_M at $P = 2, P = 3, P = 4$.
 - (d) Sketch the market demand curve, clearly showing any kink.
- ★★ Demand for restaurant meals: $Q = 200 - 5P + 0.04I$. At $P = 20, I = 50,000$:
- (a) Find equilibrium quantity.
 - (b) Compute income elasticity. Are restaurant meals normal, inferior, or luxury?
 - (c) If income rises 10%, by what percentage does quantity change approximately?
- ★★ Tea demand: $Q_T = 300 - 6P_T + 4P_C$. At $P_T = 20, P_C = 15$:
- (a) Find quantity of tea demanded.
 - (b) Compute cross-price elasticity of tea w.r.t. coffee price.
 - (c) Are tea and coffee substitutes or complements? Is this expected?
 - (d) If coffee price rises 5%, by approximately how much does tea demand change?
- ★★ Consumer demand: $P = 50 - 2Q$. Market price $P^* = \$10$.
- (a) Find Q^* .
 - (b) Compute CS at $P^* = 10$.
 - (c) Price rises to \$20. Compute new CS .
 - (d) Compute the change in CS . Decompose into rectangle and triangle.
 - (e) Sketch both situations, shading the loss in CS .
- ★★ Monopolist: $Q = 100 - 2P$ (i.e. $P = 50 - Q/2$).
- (a) Write TR as a function of Q .
 - (b) At what Q is TR maximised? What is the corresponding P ?
 - (c) What is e_d at the revenue-maximising price?

- (d) Connect your answer to (c) with the firm's optimal pricing strategy. ★ *Preview of Week 6.*

★★ [Data-Based Question] The table records price and quantity observations for two goods in the same market ($I = \$50,000$ and $P_{\text{coffee}} = \$15$ throughout unless stated).

Obs.	P_{tea}	Q_{tea}	P_{coffee}	Income I	Notes
1	\$20	160	\$15	\$50,000	Baseline
2	\$25	130	\$15	\$50,000	Price rise
3	\$20	160	\$20	\$50,000	Coffee pricier
4	\$20	200	\$15	\$60,000	Higher income
5	\$15	190	\$15	\$50,000	Price cut

- (a) Using observations 1 and 2: compute the arc price elasticity. Is tea demand elastic or inelastic? What does this imply for revenue?
- (b) Verify using observations 1 and 5 (arc elasticity). Do both estimates agree?
- (c) Using observations 1 and 3: compute the cross-price elasticity of tea w.r.t. coffee. Are they substitutes or complements?
- (d) Using observations 1 and 4: compute the income elasticity. Is tea a necessity, luxury, or inferior good?
- (e) At $P_{\text{tea}} = \$20$, the firm is considering a price rise to \$25. Using the elasticity from (a), predict the approximate change in revenue. Then verify using the actual data from observations 1 and 2.
- (f) ★ *Week 5 preview:* Using observation 1, compute consumer surplus if the demand curve is $Q = 300 - 7P$ (check: $300 - 7(20) = 160 \checkmark$) and $P^* = \$20$. What happens to CS if price rises to \$25?

★★ [Explain to a Friend — Level 2] A friend is confused about cross-price elasticity. She says: "Both printers and printer ink are expensive, and when printer prices fall, ink sales rise. So shouldn't their cross-price elasticity be positive?" Write a short paragraph correcting this reasoning and explaining what the sign of cross-price elasticity actually measures.

Level 3: Advanced and Long-Answer Problems

★★★ Gasoline: short-run $|e_d| \approx 0.2$; long-run $|e_d| \approx 0.7$.

- (a) Explain intuitively why long-run demand is more elastic.
- (b) Price rises 15%. Approximate short-run and long-run percentage changes in quantity.
- (c) Is gasoline elastic or inelastic in both runs? What does this imply for total spending?

- (d) Government imposes an excise tax raising petrol price. Will tax revenue be higher when demand is elastic or inelastic? Use the elasticity values to calculate.

★★★ Three consumers: $Q_1 = 10 - 2P$, $Q_2 = 15 - 3P$, $Q_3 = 5 - P$.

- Find each consumer's maximum willingness to pay (choke price).
- Derive the market demand curve for each price range.
- At $P = 3$: compute market quantity and market price elasticity.
- All incomes rise 20%; $e_I = +1.5$ for each consumer. How does market demand shift? New Q_M at $P = 3$?

★★★ Demand: $Q = 80 - 2P$, market price $P^* = 10$.

- Compute CS.
- Price rises to $P^* = 20$. Compute new CS and the loss.
- Decompose the loss into: (i) loss from continuing buyers; (ii) loss from buyers who exit.
- If demand were perfectly inelastic at $Q = 60$: what is the CS loss from the same price rise?

★★★ University bookstore: economics majors $Q_E = 200 - 4P$; non-economics majors $Q_N = 150 - 3P$.

- At $P = \$30$: compute Q for each group and total.
- Compute e_d for each group at $P = \$30$. Which is more elastic?
- Derive the market demand curve. Sketch it.
- Should the store raise or lower price from \$30 to maximise revenue? Use elasticities.

★★★ True/False — justify carefully:

- A demand curve with constant slope has constant price elasticity.
- If $e_d = -0.5$, a 10% price cut will increase total revenue.
- A firm can always increase revenue by raising its price.
- Cross-price elasticity of hot dogs and hot dog buns is positive.
- Large consumer surplus always implies the consumer is very wealthy.
- When two consumers form a market, market demand always lies to the right of both individual curves.

★★★ Smartphone demand: $Q = 500 - 2P + 0.5I - 3P_S$ (I in thousands; P_S = service plan

price). At $P = 200$, $I = 60$, $P_S = 30$:

- (a) Compute equilibrium quantity.
- (b) Own-price elasticity. Elastic or inelastic?
- (c) Income elasticity. Normal, inferior, or luxury?
- (d) Cross-price elasticity w.r.t. P_S . Substitutes or complements?
- (e) All three variables change simultaneously: $P \uparrow 5\%$, $I \uparrow 10\%$, $P_S \uparrow 8\%$. Approximate total % change in quantity. ★ *Connects to Week 5 tax incidence.*

★★★ Demand: $P = 80 - \frac{1}{2}Q$ (i.e. $Q = 160 - 2P$).

- (a) At what (P, Q) is demand unit elastic?
- (b) Compute CS when $P^* = 40$.
- (c) Compute TR at $P = 40$ and $P = 30$. Consistent with elasticity at $P = 40$?
- (d) A second consumer enters: $Q_2 = 80 - P$. Derive new market demand and CS at $P = 30$.
- (e) Government subsidy lowers price from \$40 to \$30: (i) gain in CS; (ii) total cost of subsidy. Is the gain larger or smaller than the cost? What does this imply?

★★★ Town with 100 identical consumers: $q_i = 12 - 2P$.

- (a) Derive market demand.
- (b) Compute market e_d at $P = 4$.
- (c) 50 new consumers enter: $q_j = 8 - P$. Derive new market demand (all consumers active).
- (d) At $P = 4$: percentage change in Q_M after new consumers enter?
- (e) Income rises 20%; $e_I = +0.8$. By what % does market demand shift? Sketch before/after demand curves.

★★★ Three perspectives on demand: $Q = 90 - 3P$, $P^* = 15$.

- (a) **Intuition:** Elastic or inelastic at $P = 15$? How does this relate to the midpoint?
- (b) **Algebra:** Compute e_d at $P = 15$. At what price is $|e_d| = 1$?
- (c) **Welfare:** Compute CS at $P = 15$. Government imposes price ceiling at $P = 10$: what happens to CS? Compute gain. ★ *Week 5 preview: what happens to the producer side?*

★★★ **[Explain to a Friend — Level 3]** Your friend is a business student studying pricing strategy. She has heard that “firms should always operate where demand is elastic.”

Write a careful 5–6 sentence explanation addressing: (a) why this rule makes sense for revenue maximisation; (b) whether the same rule holds if the firm has costs; and (c) one real-world example where a firm *appears* to operate in the inelastic region and why that might still be rational.

Full solutions follow in the solutions document. Attempt all questions independently before consulting them.